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Automated Extraction and Analysis of Timing-Based Paradata Features for Survey Quality Assurance
VALIDATION REPORT

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This document reports the validation results and describes the path that produces them: the datasets, the commands that reproduce every figure, and the acceptance criteria each run is held to. Every number below is taken from a file in `output/`, and the source file is named alongside it.

Conventions. Precision, recall and F1 are quoted to four decimal places in tables and rounded to two in prose. Counts are exact. Runtimes are wall clock seconds for a single named run, and the run is always named.

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0. Headline results

The authoritative evaluation is the July 2026 robustness round. It uses seeds that were never touched during development, a stress difficulty in which normal and anomalous profiles overlap by construction, and a class imbalance sweep. Detector settings are the defaults, with nothing fitted to the evaluated data.

Measure	Result	Proposal threshold
Precision, across all runs	0.96 to 1.00	0.80
Recall, across all runs	0.84 to 1.00	0.75

The lowest single precision observed is 0.9570, on seed 7 in ODK format at stress difficulty. The lowest recall is 0.84, on seed 99 in ODK format.

Per-class recall by anomaly type. Each figure is the recall for that injected class, that is the proportion of interviews of that type which the detector flagged, aggregated across all four stress runs rather than quoted from any single run. These are diagnostic breakdowns of the combined detector's recall, not separate detectors and not separate success criteria.

Anomaly type	Detected	Per-class recall	Per-run range
Speeding	132 of 132	1.00	1.00 to 1.00
Disengaged	121 of 136	0.89	0.82 to 0.94
Fabricated	93 of 132	0.70	0.61 to 0.76

The proposal's success criteria, precision at or above 0.80 and recall at or above 0.75, are defined for the combined detector and are met on every run with margin, as the table above shows. No committed target is missed. The per-class figures below that threshold, in particular fabricated recall at 0.70, are reported because they locate where the method is weakest, not because they breach a commitment.

Fabrication is the weakest of the three and is discussed in section 5. The entire round is reproducible with one command, which regenerates its datasets from recorded seeds and evaluates every condition:

```
python scripts/run_robustness_validation.py
```

Methodology and full tables: `output/robustness_validation/robustness_summary.md`.

0.1 Auditable evidence

Validated state. Tag `v-validation-complete`. The exact commit it resolves to is verifiable with:

```
git rev-parse v-validation-complete
```

The tag table at the end of `docs/evidence_index.md` records the same mapping and what is present in that tree.

Test suite. Command and output, run on Python 3.14.0 with pandas 3.0.1:

```
$ python -m pytest -q
....................................
[100%]
37 passed in 5.99s
```

Feature coverage. `docs/feature_coverage.md` tabulates all 40 catalogued features across the Q, S, V and T families, showing where each is implemented, whether the pipeline exported it in the recorded canonical run, and how it is covered by tests. Summary: 40 of 40 implemented, 40 of 40 exported, 34 of 40 covered by a direct feature-naming test, 40 of 40 exercised end to end. Every one of the 30 features the validity analysis found significant now has a direct test; the six without one are constant across the synthetic corpus. The table is regenerated by `python scripts/build_feature_coverage.py` rather than maintained by hand.

Granular outputs. The question-level export carries all ten Q features, the section-level export all ten S features, and the interview-level feature dataset all forty, ten in each of the four families. Survey-level (V) and statistical (T) outputs are complete.

Confusion counts, per run.

Run	Format	TP	FN	FP	TN	Precision	Recall	F1
`benchmark_n500_odk`	ODK	100	0	0	400	1.0000	1.0000	1.0000
`benchmark_n500_surveycto`	SurveyCTO	100	0	0	400	1.0000	1.0000	1.0000
`stress_seed7_odk`	ODK	89	11	4	396	0.9570	0.8900	0.9223
`stress_seed7_surveycto`	SurveyCTO	85	15	2	398	0.9770	0.8500	0.9091
`stress_seed99_odk`	ODK	84	16	0	400	1.0000	0.8400	0.9130
`stress_seed99_surveycto`	SurveyCTO	88	12	0	400	1.0000	0.8800	0.9362
`sweep_r005`	SurveyCTO	25	0	1	474	0.9615	1.0000	0.9804

Run	Format	TP	FN	FP	TN	Precision	Recall	F1
`sweep_r010`	SurveyCTO	50	0	0	450	1.0000	1.0000	1.0000
`sweep_r020`	SurveyCTO	100	0	0	400	1.0000	1.0000	1.0000
`sweep_r030`	SurveyCTO	150	0	0	350	1.0000	1.0000	1.0000

Confusion counts, aggregated by platform format.

Format	TP	FN	FP	TN	Precision	Recall
ODK	273	27	4	1,196	0.9856	0.9100
SurveyCTO	598	27	3	2,872	0.9950	0.9568

Both formats clear the proposal criteria. ODK carries all four false positives across the whole round and a slightly lower recall, which is consistent with its timestamps arriving as ISO strings rather than epoch milliseconds and therefore passing through one more conversion step. The difference is four interviews in 1,500 and does not affect any verdict.

Per-class recall by anomaly type is in section 0 above, and the diagnostic behind the fabricated figure is `docs/fabricated_detection_diagnostic.md`. **Baselines** are section 2, **effect sizes** section 1, and the **class-imbalance sensitivity check** section 4.

1. Feature validity and effect sizes

Each of the 40 catalogued features was tested for its ability to separate normal from anomalous interviews, using Mann-Whitney U on the n500 SurveyCTO benchmark (400 normal, 100 anomalous) at alpha 0.05, with p-values adjusted for multiple comparisons. Cohen's d is the primary effect size, with a rank-biserial correlation derived from Mann-Whitney U reported alongside it.

30 of the 40 features are significant. The largest effects are concentrated in the statistical and timing families:

Feature	Cohen's d	Adjusted p
T-03 response time entropy	-10.57	1.6e-52

Feature	Cohen's d	Adjusted p
Q-10 first to last interaction duration	-4.14	3.0e-52
T-06 response time mean	-4.14	3.0e-52
S-01 mean section pace	-4.14	3.0e-52
Q-01 response time	-4.14	3.0e-52
S-02 section duration	-3.80	8.2e-52
V-01 total interview duration	-3.79	1.1e-51
V-09 session length vs median	-3.79	1.1e-51
T-02 response time IQR	-3.51	2.0e-52
T-01 response time variance	-1.87	2.0e-52

The ten features that do not discriminate are V-07 time of day, V-08 day of week, S-04 section to section pace ratio, S-05 and S-06 pause counts, S-09 section back navigation count, S-10 section completion rate, and Q-04 to Q-06 revisit flag, revisit count and field completion ratio. The first two are contextual rather than behavioural. The remainder are constant across the synthetic corpus and therefore carry no signal in it. All ten are retained in the output for analysts but carry no weight in detection.

Full table, including rank-biserial values, group means and standard deviations: `output/feature_validation_results.csv` and `output/feature_validation_results.txt`.

2. Baseline comparison

The proposal requires the full detector to be compared against two methods representing current practice. Both baselines are recomputed from the raw files, the generated features and the ground truth labels inside `scripts/strict_examiner_audit.py`, so the comparison needs no manual work.

On the n500 SurveyCTO benchmark, 100 anomalies in 500 interviews:

Method	Precision	Recall	F1

Method	Precision	Recall	F1
Full detector	1.0000	1.0000	1.0000
Duration-only IQR baseline	0.9565	0.6600	0.7811
Zero back navigation baseline	0.4648	0.6600	0.5455

The duration-only baseline reaches acceptable precision but misses 34 of the 100 anomalies, every one of which has a total duration inside the Tukey fences. The zero back navigation baseline flags 142 interviews to find the same 66, giving 76 false positives. Neither baseline meets the proposal's recall threshold of 0.75. This is the central result of the project: aggregate duration alone misses about a third of the injected anomalies that multi-feature analysis captures.

Source: `output/examiner_audit/strict_examiner_audit_report.md`.

2.1 Contribution of each detection component

The proposal requires separate performance breakdowns for the IQR-based statistical component and the rule-based component, so that the contribution of each can be assessed rather than only their combined effect.

The combined detector flags an interview when `anomaly_score` reaches the behaviour threshold of 2, where `anomaly_score = min(iqr_flags, 1) + rule_flags`. The IQR contribution is capped at one on purpose, so that several correlated duration, mean and variance outliers arising from the same underlying timing pattern are not counted repeatedly. That cap makes the threshold of 2 unreachable for the IQR component alone, so each component is isolated as follows:

- **IQR only:** any IQR fence fired, `iqr_flags >= 1`.
- **Rules only:** `rule_flags >= 2`, the same threshold the combined detector applies, with the IQR contribution removed.
- **Combined:** the shipped detector, unchanged.

Aggregated over all ten runs of the robustness round:

Component	Precision	Recall	F1	TP	FP	FN

Component	Precision	Recall	F1	TP	FP	FN
Combined	0.9920	0.9416	0.9662	871	7	54
Rules only	0.9988	0.9319	0.9642	862	1	63
IQR only	0.6628	0.6162	0.6387	570	290	355

Per run:

Run	Combined P / R	Rules only P / R	IQR only P / R
`benchmark_n500_odk`	1.0000 / 1.0000	1.0000 / 1.0000	0.8547 / 1.0000
`benchmark_n500_surveycto`	1.0000 / 1.0000	1.0000 / 1.0000	0.8929 / 1.0000
`stress_seed7_odk`	0.9570 / 0.8900	0.9884 / 0.8500	0.3784 / 0.2800
`stress_seed7_surveycto`	0.9770 / 0.8500	1.0000 / 0.8100	0.3158 / 0.2400
`stress_seed99_odk`	1.0000 / 0.8400	1.0000 / 0.8300	0.3167 / 0.1900
`stress_seed99_surveycto`	1.0000 / 0.8800	1.0000 / 0.8800	0.3200 / 0.2400
`sweep_r005`	0.9615 / 1.0000	1.0000 / 1.0000	0.4717 / 1.0000
`sweep_r010`	1.0000 / 1.0000	1.0000 / 1.0000	0.6667 / 1.0000
`sweep_r020`	1.0000 / 1.0000	1.0000 / 1.0000	0.9174 / 1.0000
`sweep_r030`	1.0000 / 1.0000	1.0000 / 1.0000	0.9174 / 0.6667

What the breakdown shows

The rule layer carries the detection. On its own it reaches precision 0.9988 and recall 0.9319, which is within a point of the combined detector on both measures. The IQR layer on its own reaches precision 0.6628 and recall 0.6162, and under the stress condition its recall falls to between 0.19 and 0.28.

Combining the two is nonetheless worth doing, but the margin is small and should be stated as such: the combination recovers nine true positives that the rules miss, lifting recall from 0.9319 to 0.9416, and costs six additional false positives, moving precision from 0.9988 to 0.9920. That is the entire measured contribution of the IQR component in the shipped configuration.

The IQR component also degrades in a specific and instructive way. Across the class imbalance sweep its recall holds at 1.00 while anomalies make up 5, 10 and 20 percent of the sample, then falls to 0.6667 at 30 percent. This is masking: the fences are estimated from a sample that contains the outliers they are meant to isolate, so as contamination rises the quartiles move with it and the fences stop reaching the anomalies. The same mechanism accounts for part of the fabricated recall shortfall analysed in `docs/fabricated_detection_diagnostic.md`, and it is a property of applying unsupervised fences to contaminated data rather than a defect in the implementation.

This breakdown is regenerated from the recorded predictions, not from a fresh run, with:

```
python scripts/component_breakdown.py
```

Full per-run values including confusion counts: `output/robustness_validation/component_breakdown.csv`.

3. Canonical validation set

The canonical labelled run is the contract test. It uses clean, reproducible profiles and is deliberately easy, so it verifies the intended mapping between profile and category rather than proving field performance.

```
python run_pipeline.py \
  --inputs data/workflow_validation/canonical_validation/odk.csv
data/workflow_validation/canonical_validation/surveycto.csv \
  --parser auto \
  --output-dir output \
  --label canonical_validation \
  --ground-truth
data/workflow_validation/canonical_validation/ground_truth_labels.csv
```

Verified result: precision 1.0000, recall 1.0000, F1 1.0000.

The run writes `output/question_features_canonical_validation.csv`, `output/section_features_canonical_validation.csv`, `output/feature_dataset_canonical_validation.csv`,

`output/anomaly_report_canonical_validation.csv`, `output/pipeline_output_canonical_validation.csv`, `output/predictions_canonical_validation.csv` and `output/metrics_canonical_validation.{csv,json,md}`.

4. Class imbalance sweep

Held-out seed 7, canonical difficulty, SurveyCTO, 500 interviews, with the proportion of anomalies varied.

Anomaly prevalence	Precision	Recall	F1
5 percent	0.9615	1.0000	0.9804
10 percent	1.0000	1.0000	1.0000
20 percent	1.0000	1.0000	1.0000
30 percent	1.0000	1.0000	1.0000

Recall holds at 1.00 throughout. Precision falls only at 5 percent prevalence, where a single false positive among few true anomalies costs almost four points. This is a property of the metric rather than of the detector, but it matters operationally, because real fabrication rates are usually low and the figure quantifies the review burden a deployment should expect.

Source: `output/robustness_validation/metrics_sweep_r*.csv`.

5. Held-out stress evaluation

The stress generator overlaps the profiles on purpose. Some genuine respondents are fast with no back navigation, and some fabricated interviews are only mildly compressed. Seeds 7 and 99 were never used to develop or calibrate anything.

Dataset	Precision	Recall	F1
Seed 7, SurveyCTO	0.9770	0.8500	0.9091
Seed 7, ODK	0.9570	0.8900	0.9223
Seed 99, SurveyCTO	1.0000	0.8800	0.9362

Dataset	Precision	Recall	F1
Seed 99, ODK	1.0000	0.8400	0.9130

All four runs clear both success criteria, and performance is consistent across the two platform formats. That consistency is itself a result: the same detector, unmodified, performs equivalently on structurally different inputs, which is what the common intermediate representation was designed to achieve.

Per-class recall by anomaly type, all four runs. Each cell is the count of that injected class the detector flagged, over the count present:

Run	Speeding	Disengaged	Fabricated
Seed 7, SurveyCTO	33/33	28/34	24/33
Seed 7, ODK	33/33	32/34	24/33
Seed 99, SurveyCTO	33/33	30/34	25/33
Seed 99, ODK	33/33	31/34	20/33
Aggregate	**132/132 (1.00)**	**121/136 (0.89)**	**93/132 (0.70)**

What the misses are

All undetected fabricated interviews come from the stress generator's mildly compressed variant: mean response time around 3,800 ms, moderate spread, one or two back navigations. The generator emits two fabricated profiles under stress difficulty, and the tightly compressed one is detected without exception.

A bounded diagnostic pass on these false negatives, run on development seeds 11 and 42 and reported in `docs/fabricated_detection_diagnostic.md`, established that they are **not** indistinguishable from genuine interviews. Response time entropy separates the two populations completely on those datasets: the highest fabricated entropy observed is 1.983 and the lowest genuine entropy is 2.039, with no overlap. The misses arise from how the detector reads that feature rather than from an absence of signal in it. Two mechanisms are responsible. The behavioural entropy rule uses an absolute constant of 0.5, which is calibrated to the tightly compressed profile and cannot fire for the mildly compressed one. The entropy IQR fence is

computed on a dataset containing 20 percent injected anomalies, which pulls the lower quartile down and the fence with it, so it fires for a minority of the misses where a fence estimated from genuine interviews alone would separate all of them.

A related structural point applies to the other statistical features. For strongly right-skewed non-negative quantities such as response time variance and IQR, the Tukey lower fence `Q1 - 1.5 x IQR` falls below zero and is therefore unreachable, so no low-side outlier can be detected on those features at all. Entropy is the exception because it is bounded and roughly symmetric.

This corrects an earlier explanation in this document, which attributed the misses to a joint pattern that univariate methods cannot capture. The data does not support that reading. No threshold was changed in producing the diagnostic and the results reported above stand unaltered.

The correction exists, and is deliberately not in the evaluated system

The diagnosis was acted on. A corrected entropy fence was developed on the same development seeds, frozen, and evaluated once against the held-out data. It keeps the absolute constant as a floor and adds a fence estimated from the uncontaminated upper half of the entropy distribution, a form chosen because every anomaly profile in scope lowers entropy and none raises it, and because it reduces to the ordinary Tukey fence for a symmetric distribution. Under that correction, fabricated per-class recall rises from 0.70 to 1.00 and the lowest precision observed anywhere in the round falls from 0.9570 to 0.8333.

The gain concentrates in the stress condition and the cost concentrates at low prevalence, where recall was already complete and the extra flags fall entirely on genuine interviews. Since realistic fabrication rates are low, the correction as it stands buys no additional detection in the condition that most resembles deployment and pays for it in false accusations. It is therefore held on a separate branch and is not part of the system evaluated in this document. Every figure in this document describes the shipped detector without that change.

The full trade-off, and the argument that adopting it is a deployment judgement about the relative cost of an undetected fabrication against a genuine enumerator sent for review rather than a methodological question, is set out in section 7 of `report/final_academic_report.md`.

Source: `output/robustness_validation/metrics_stress_seed*.csv` and `detection_by_type_stress_seed*.csv`.

6. Earlier stress sweep, seeds 11 and 42

A separate stress sweep on seeds 11 and 42 predates the held-out round and is retained for continuity. It is not the authoritative result; section 5 is.

```
python scripts/validate_production_readiness.py --sizes 500 --seeds 11 42 --formats surveycto odk --difficulty stress --data-root data/stress_benchmarks --output-dir output/stress_validation
```

It shows precision between 0.9770 and 0.9890 and recall between 0.8000 and 0.9000, clearing both thresholds. It does not pass the strict score contract: some subtle fabricated interviews score as REVIEW rather than reaching 5, and a few intentionally borderline anomalies remain CLEAN. The canonical datasets verify the proposal contract; stress mode maps where robustness has limits.

Source: `output/stress_validation/production_readiness_summary.md`.

7. Larger benchmark packs and scale

The smaller labelled synthetic packs live in `data/synthetic_datasets/n500` and `n2000`. The larger `n10000` pack is regenerated from its recorded seed before rerunning the scale check:

```
python tests/generate_benchmark_datasets.py --dataset-size 10000 --seed 42 --output-dir data/synthetic_datasets/n10000
```

```
python scripts/analyze_pipeline_metrics.py --datasets n500 n2000 n10000 --formats surveycto --output-dir output
```

Cross-format spot-check:

```
python scripts/analyze_pipeline_metrics.py --datasets n500 --formats odk --output-dir output
```

Cross-format parity at n500 is exact: SurveyCTO 1.0000/1.0000 and ODK 1.0000/1.0000. Scaling from 500 to 10,000 interviews, precision moves from 1.0000 to 0.9945 while recall holds at 1.0000.

Source of the scaling figures

The runtime figures quoted here and in the final report are those of the strict examiner audit, which is the run that carries the scale check as an explicit pass or fail condition. Its raw times are:

Dataset	Format	Interviews	Runtime	Precision	Recall
`n500`	SurveyCTO	500	56.12 s	1.0000	1.0000
`n10000`	SurveyCTO	10,000	1,068.87 s	0.9945	1.0000

A twentyfold increase in input costs a factor of 19.05 in runtime, which is consistent with the near-linear complexity the design targets.

Source: `output/examiner_audit/strict_examiner_audit_report.md` and its JSON companion, in which the same two times appear as `n500_runtime_seconds` and `n10000_runtime_seconds` under the scale stability check.

These are wall clock timings of one run and they move between runs on the same machine. A separate aggregate timing file records 57.54 s and 1,037.74 s for the same two datasets, a factor of 18.04, from a different run: `output/synthetic_dataset_metrics.csv` and its JSON companion. The two pairs are not one measurement reported twice and should not be read as such. The strict examiner audit is the source cited throughout this document, and both runs place the runtime factor below the twentyfold growth in input size.

Aggregate outputs: `output/synthetic_dataset_metrics.csv`, `output/synthetic_dataset_metrics.md`, `output/synthetic_detection_by_type.csv`.

8. Production readiness sweep

To confirm the detector still meets the proposal contract on newly generated data:

```
python scripts/validate_production_readiness.py --sizes 120 500 --seeds 11 42
--formats surveycto odk
```

Each run is checked against precision at or above 0.80, recall at or above 0.75, fabricated interviews scoring at least 5, and normal interviews scoring at most

1. Summaries are written to `output/production_validation/production_readiness_summary.{csv,json,md}`.

9. Reproducibility

Full workflow, which generates its own labelled validation bundle and does not overwrite user-supplied files under `data/`:

```
python scripts/run_complete_workflow.py
```

Strict end-to-end audit:

```
python scripts/strict_examiner_audit.py
```

The examiner audit checks raw-format consistency against the format guide, CIR normalisation, manual recalculation of selected features, anomaly score reconstruction, reproducibility of precision and recall from ground truth, fabricated versus normal feature separation, baseline comparisons, SurveyCTO and ODK parity, and scale behaviour between n500 and n10000. Every check passes in the recorded run.

Test suite: 37 tests, all passing, verified on Python 3.14.0 with pandas 3.0.1.

```
python -m pytest
```